ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

Biographical Note: Sir Tim Berners-Lee

Sir Timothy John Berners-Lee is a British computer scientist, visionary technologist, and open-web advocate who invented the World Wide Web, reshaping how humanity shares knowledge, communicates, and does business. Born on 8 June 1955 in London, he grew up in a household where computing was part of everyday conversation: both his parents, Conway Berners-Lee and Mary Lee Woods, worked on the Ferranti Mark 1, one of the first commercially built computers. Berners-Lee's foundational belief is that the web should be a universal, open space that belongs to everyone, not a product controlled by any single company or government.

He studied at Emanuel School in London and then at The Queen's College, Oxford, where he graduated with a first-class degree in physics in 1976. As a student he famously built his own computer from a soldering iron, TTL gates, an M6800 processor, and an old television. After graduating, he worked as an engineer at Plessey Telecommunications and later at D.G. Nash Ltd, developing typesetting software and real-time systems, which built the practical engineering foundation for his later work.

In 1980, Berners-Lee took a short contract as a software consultant at CERN, the European particle physics laboratory near Geneva. Frustrated by how hard it was for researchers to find information across incompatible computers and systems, he wrote a personal program called ENQUIRE, which linked pieces of information together in a web-like structure. He returned to CERN as a fellow in 1984, and in March 1989 he submitted a proposal titled "Information Management: A Proposal", describing a distributed hypertext system that would connect documents across the internet. His supervisor Mike Sendall described the idea as "vague but exciting", and allowed him to pursue it.

Working with Belgian engineer Robert Cailliau, Berners-Lee turned the idea into reality on a NeXT computer in 1990. He created the three core technologies that still underpin the web today: the Uniform Resource Locator (URL) for addressing resources, the HyperText Transfer Protocol (HTTP) for transferring them, and the HyperText Markup Language (HTML) for describing pages. He also wrote the first web browser and editor, called WorldWideWeb, and the first web server. The first website, hosted at CERN, explained the project itself, and the web was opened to the wider world in 1991.

A defining decision came on 30 April 1993, when CERN placed the World Wide Web software in the public domain, making it free for anyone to use with no royalties. Berners-Lee has consistently said that this openness was essential to the web's explosive growth. In 1994 he moved to the Massachusetts Institute of Technology and founded the World Wide Web Consortium (W3C), an international body that develops open standards to keep the web interoperable, accessible, and evolving. He continues to serve as its Director, and he also holds professorships at MIT and the University of Oxford.

Berners-Lee's vision extends beyond linked documents. He has championed the Semantic Web, an effort to make data on the web machine-readable so that computers can understand and connect information more intelligently. In 2009 he founded the World Wide Web Foundation to advance the open web as a public good and a basic right, and in 2012 he co-founded the Open Data Institute with Nigel Shadbolt to promote open data. More recently, concerned about data misuse and the concentration of power online, he launched the Solid project, which lets people store personal data in their own data "pods" and control who can access it, along with the company Inrupt to support its adoption. In 2018 he also launched the Contract for the Web, which calls on governments, companies, and citizens to protect the web's openness and safety.

His contributions have earned him major recognition. He was knighted by Queen Elizabeth II in 2004 and received the Millennium Technology Prize that same year. In 2007 he was appointed to the Order of Merit, and in 2016 he received the ACM A.M. Turing Award, often called the Nobel Prize of computing. At the opening ceremony of the 2012 London Olympics, he appeared live with a message that summed up his philosophy: "This is for everyone." He has also shared his story in books such as Weaving the Web (1999).

Through his inventions, his decision to keep the web free and open, and his ongoing advocacy for privacy, access, and a healthier digital world, Tim Berners-Lee remains one of the most influential figures in the history of technology. His journey shows how a simple idea, born from a practical problem and shared generously, can connect billions of people across the globe.

Contact

www.linkedin.com/in/hanu-shree-hanu-ab068642b (LinkedIn)

github.com/hanushree-cse/hanushree_cse (GitHub)

Top Skills

C

Python (Programming Language)

Git & GitHub

Problem Solving

HANU SHREE K.B

First-Year CSE Student (Batch 2026–2030) | Aspiring Software Developer | C & Python

Chennai, Tamil Nadu, India

Summary

" Good software starts with strong fundamentals, curiosity, and the habit of solving real problems ".

I am a first-year Computer Science and Engineering (CSE) student at St. Joseph's College of Engineering, Chennai. I am building my foundation in C and Python programming and exploring software development, problem solving, and technology-driven solutions. I am particularly interested in improving my coding skills and understanding how software can be used to solve real-world problems.

Along with academics, I have participated in hackathons and technical activities, which have helped me develop creativity, teamwork, and a problem-solving mindset. I am also building my technical profile through GitHub and LinkedIn. I am a hardworking, quick learner, always willing to learn new technologies and take on new challenges.

Key Areas of Interest:

- **Software Development:** writing clean, structured programs and learning how applications are built.
- **Problem Solving:** practising logic and algorithmic thinking to break problems into clear steps.
- **Technology-Driven Solutions:** applying code to create practical, meaningful impact.
- **Open-Source & Version Control:** growing a public portfolio with Git and GitHub.

Tech Stack & Focus Tools:

- **Languages:** C, Python (Core Syntax, Data Structuring)
- **Core Tools:** Git, GitHub, VS Code
- **Strengths:** Quick learner, teamwork, creativity, adaptability

Experience

St. Joseph's College Of Engineering
Student
August 2026 - Present
Chennai

First-year undergraduate student pursuing a Bachelor of Engineering (B.E.) in Computer Science and Engineering. Currently building programming fundamentals in C and Python while developing structured coding workflows and problem-solving skills.

Hackathons & Technical Activities
Participant

Took part in hackathons and technical activities, developing creativity, teamwork, and a problem-solving mindset.

Education

St. Joseph's College Of Engineering
COMPUTER SCIENCE AND ENGINEERING · (2026)

St. Joseph's College Of Engineering
Bachelor of Engineering, Computer Science and Engineering · (August 2026 - May 2030)

Career Goals

Short term: Strengthen programming fundamentals in C and Python, build projects, and grow a strong GitHub portfolio.

Long term: Become a confident software professional who continuously improves and builds meaningful technology solutions that create real-world impact.

SWOC ANALYSIS

My Software Engineering Roadmap to 2030

Hanu Shree K.B | First-Year B.E. CSE, St. Joseph's College of Engineering, Chennai

PROFILE & CAREER VISION

- **Who I am:** First-year CSE student learning C and Python; interested in software development and problem solving; taken part in hackathons and technical activities.
- **Vision:** Become a confident software professional who builds technology solutions with real-world impact.
- **Objective:** Turn a strong C and Python foundation into real projects, deeper problem solving and industry-ready skills.
- **Framework:** Use the SWOC matrix to find my gaps and trends, then convert them into a year-by-year plan.

STRENGTHS

- **Building a strong foundation:** Learning the basics of Python and C step by step, the base for every advanced topic.
- **Eager to gain knowledge:** Keen to keep learning, which matters most in a field that changes fast.
- **Two complementary languages:** C builds understanding of how computers work; Python makes building ideas quick.
- **Hackathon and tech exposure:** Taken part in hackathons and technical activities, so I am used to learning by doing.

WEAKNESSES

- **Still in my basic stage:** At the beginning of the journey; need to upgrade my skills steadily.
- **Limited hands-on depth:** Need more projects and practice to move from knowing concepts to applying them.
- **Core computer science to build:** Data structures, algorithms and problem solving need regular, focused practice.
- **No public body of work yet:** A growing GitHub portfolio and clear project write-ups will make my skills visible.

OPPORTUNITIES BY 2030

- **AI-assisted development:** Engineers who build with and around AI tools will be in high demand.
- **Growing tech fields:** Cloud, cybersecurity, data engineering and IoT all need strong programmers; C fits embedded work well.
- **Remote and global work:** Many companies now hire across borders, widening the job market.
- **Open source and startups:** GitHub contributions and hackathon projects can open doors early.
- **Digital growth in India:** Healthcare, fintech, education and governance need new software solutions.
- **Learning ecosystem:** Free courses, student programs, communities and competitions make skill-building affordable.

CHALLENGES BY 2030

- **Fast-changing technology:** Tools and languages shift quickly, so continuous learning is a must.
- **Coding automation:** Simple tasks may be handled by AI, so I must stand out through problem solving and design thinking.
- **Heavy competition:** A large pool of engineering graduates means projects and practical skills matter beyond marks.
- **Industry skill gap:** Classroom learning alone may not match what companies expect.
- **Depth versus breadth:** Choosing what to specialize in without spreading too thin.
- **Consistency over time:** Staying disciplined and motivated across four years of study.

TURNING SWOC INTO STRATEGY

- **Strength + Opportunity:** Use my Python and C base to explore AI, cloud and embedded projects, and share them on GitHub.
- **Strength + Challenge:** Use my eagerness to learn to keep pace with changing tools and stand out from routine coding.
- **Weakness + Opportunity:** Use free courses and communities to fix basics and gain project experience quickly.
- **Weakness + Challenge:** Avoid shallow learning: practise problem solving daily so competition does not expose gaps.

ACTION PLAN

- **Daily practice:** Solve a few Python and C problems every day, increasing difficulty.
- **Build small projects:** Finish one small project each month, such as a calculator, a to-do app or a game.
- **Learn core CS:** Study data structures and algorithms alongside coursework.
- **Show my work:** Keep a clean GitHub with a README for every project and update LinkedIn.
- **Use AI as a tool:** Let it speed up routine work, but learn to explain every line I write.
- **Join hackathons / open source:** Keep participating in hackathons; start with small open-source fixes and documentation.

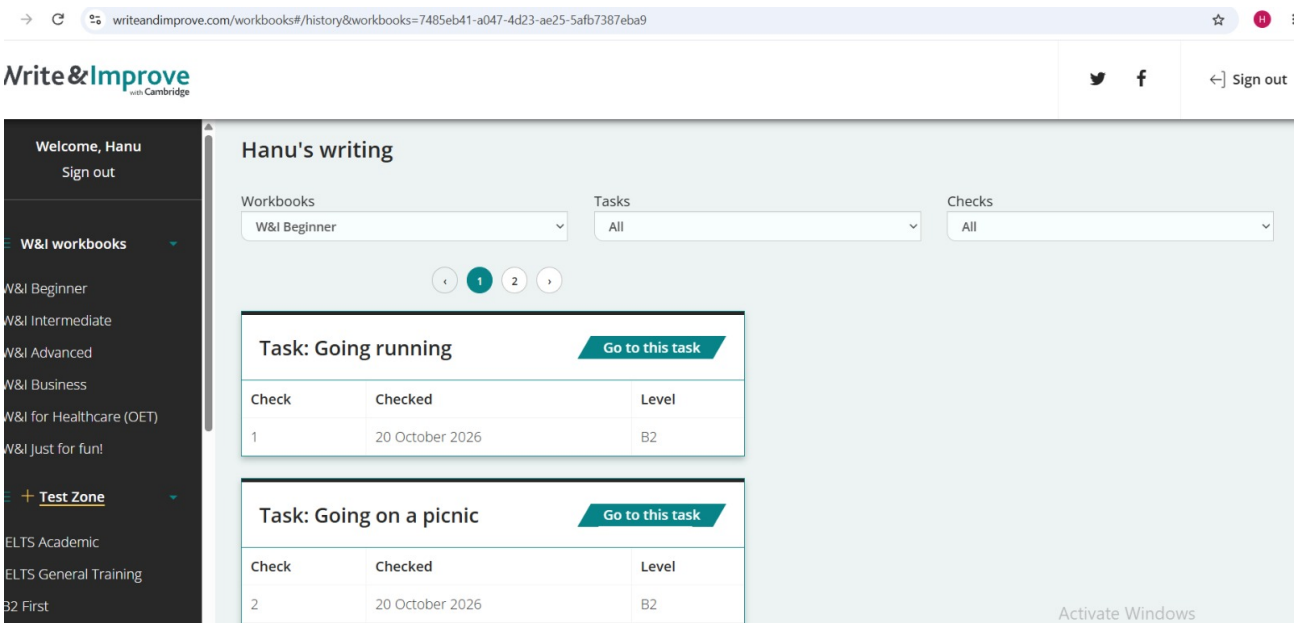
4-YEAR MILESTONE ROADMAP (2026 - 2030)

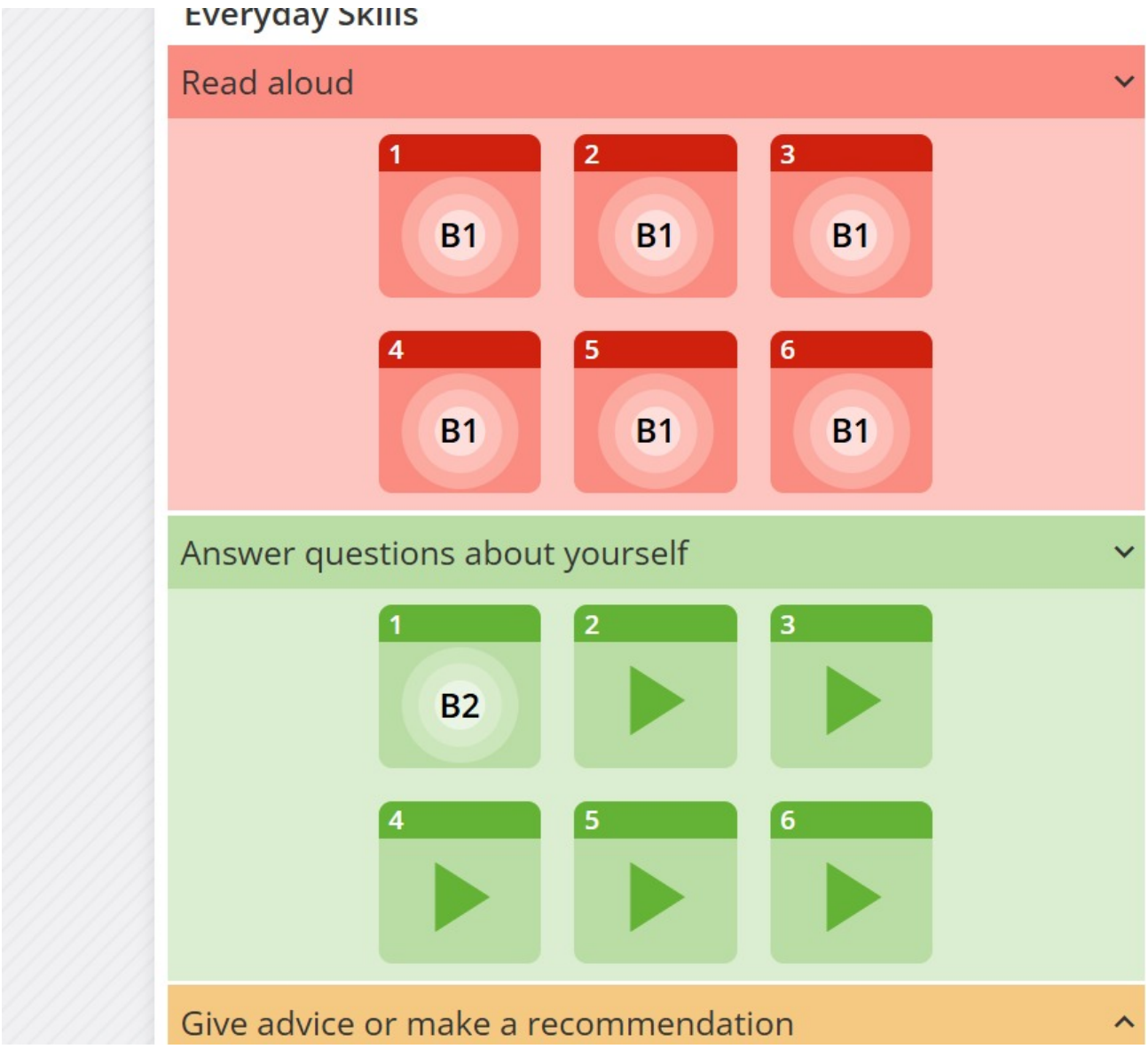
Year	Primary Focus	Target Deliverable
Year 1 (2026-27)	Python, C, problem solving	3 small projects on GitHub
Year 2 (2027-28)	Data structures, algorithms, web basics	Full-stack mini project
Year 3 (2028-29)	Specialization (AI, cloud or embedded)	Capstone-style project + hackathon
Year 4 (2029-30)	Interview prep, advanced projects	Portfolio + final-year project

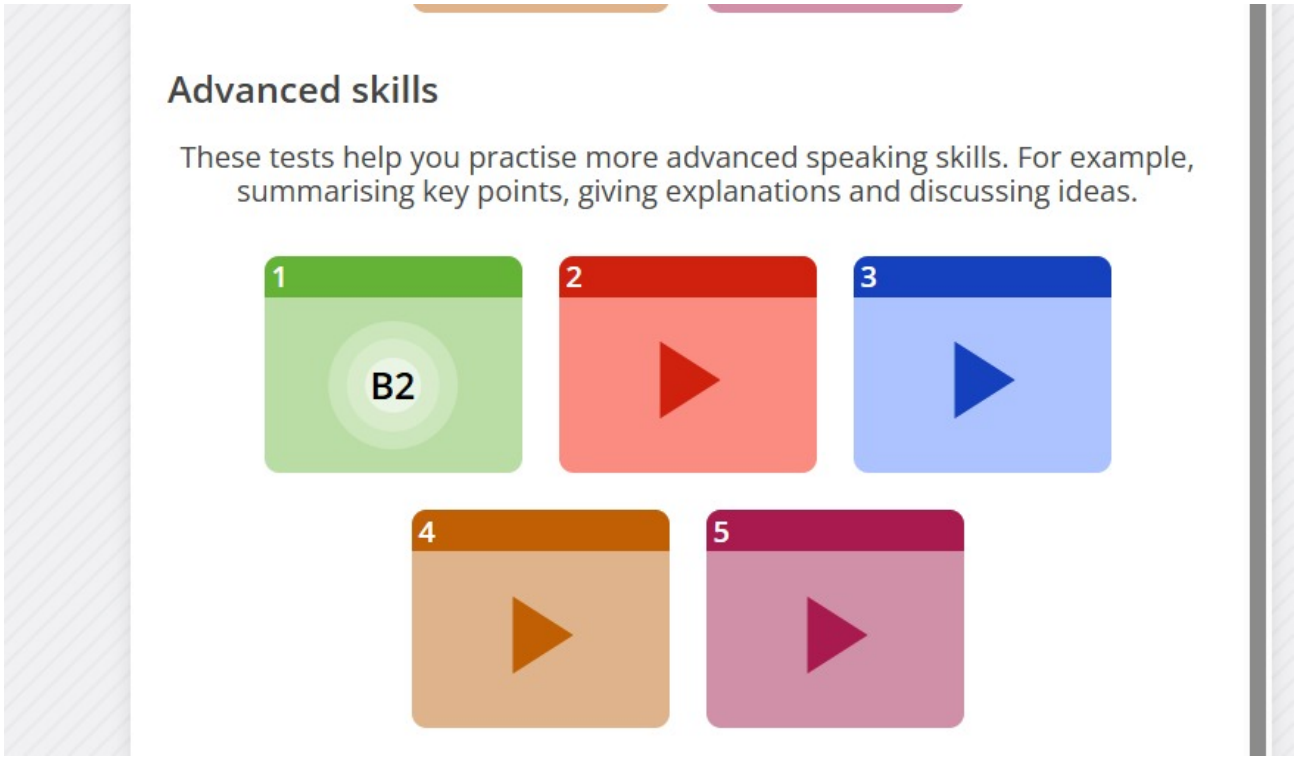
CONCLUSION

Start strong with my Python and C foundation and eagerness to learn; grow steadily by closing my basic-stage gaps through daily practice and projects; aim for 2030 by riding trends in AI, cloud and software to build technology with real-world impact.

Strong basics today, real-world impact by 2030.


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W&I for Healthcare (OET)
W&I Just for fun!
Test Zone
ELTS Academic
ELTS General Training
B2 First
Hanu's writing
Workbooks
W&I Beginner
Tasks
All
Checks
All
1
2
Task: Going running
Go to this task
Check
Checked
Level
1
20 October 2026
B2
Task: Going on a picnic
Go to this task
Check
Checked
Level
2
20 October 2026
B2
Activate Windows


Everyday Skills
Read aloud
1
B1
2
B1
3
B1
4
B1
5
B1
6
B1
Answer questions about yourself
1
B2
2
3
4
5
6
Give advice or make a recommendation


Advanced skills
These tests help you practise more advanced speaking skills. For example, summarising key points, giving explanations and discussing ideas.
1
B2
2
3
4
5

ENHANCING SOFT SKILLS AND PERSONALITY

8-Week NPTEL Course • Rewritten Course Summary

Instructor: Prof. T. Ravichandran, IIT Kanpur

Course Overview
This revised summary retains the complete learning coverage of the original 8-week course while presenting the ideas in fresh wording and a cleaner academic format.

Week 1: Building Soft Skills: Perception, Self-Awareness & Personal Growth

The opening week establishes soft-skill development as a process of real behavioural change rather than a superficial exercise or a way of collecting certificates. The course presents personal development as a holistic journey involving physical, mental, emotional, psychological, social, cultural and spiritual dimensions. It also stresses the importance of human relationships, observing that they have a major influence on personal happiness.

Stephen Covey's idea of "beginning with the end in mind" is used to encourage learners to compare everyday actions with long-term goals and remove habits that do not contribute value. The discussion on perception, including the story of the blind men and the elephant, demonstrates how the same situation can be understood differently because of personal assumptions and prejudice. The principle "seek first to understand, and then to be understood" is introduced as a practical foundation for empathy.

The week then focuses on nine self-management abilities: self-awareness, self-confidence, mindset, emotional balance, stress handling, coping with failure, patience and trustworthiness, perceptiveness, and spiritual intelligence. Emotional stability combined with spiritual intelligence is linked with stronger personal credibility. The contrast between the "law of abundance" and "poverty of mind" explains how optimistic thinking encourages courage and fulfilment, whereas scarcity-based thinking encourages fear and defensiveness. Maslow's hierarchy highlights self-actualization as the highest level of human growth, while McClelland's achievement motivation theory and Danah Zohar's principles of spiritual quotient (SQ) encourage a meaningful, value-oriented approach to life.

Week 2: Managing Conflict, Overcoming Procrastination & Practising Assertiveness

Week 2 presents conflict as a normal result of differences in goals, opinions or expectations and focuses on handling it constructively instead of avoiding it. Proper communication can turn conflict into an opportunity for productivity and stronger relationships, while poor communication can intensify disagreement. Negotiation, mediation and arbitration are discussed with emphasis on solutions that create a sustainable win-win outcome and preserve goodwill.

Case examples involving family and interpersonal disagreements show why both perspectives must be examined and why hidden needs or a possible "third side" should not be ignored. Effective conflict resolution

therefore requires identifying the deeper human needs behind a demand rather than concentrating only on rigid surface-level solutions. The module identifies three forms of conflict: intrapersonal, interpersonal and intergroup, with intrapersonal conflict described as particularly difficult because the struggle occurs within oneself.

The week also examines procrastination and introduces 13 practical methods for overcoming it. Brian Tracy's "eat that frog" and "eat the elephant bite by bite" ideas encourage learners to begin with difficult priorities and divide large tasks into smaller actions. Kaizen is presented as a way to overcome resistance by committing to a very small amount of work, such as one minute at a fixed time each day. Clean workspaces, public commitment and self-reward can further strengthen productivity.

The final focus is assertiveness. Learners are warned against the "compliance trap" of saying yes merely to satisfy others or avoid discomfort. Assertiveness means expressing one's views and protecting one's rights while also respecting the rights of other people.

Week 3: Habit Development, Behavioural Loops & Cognitive Productivity

Week 3 explores habits as largely automatic behavioural responses that are activated by particular thoughts, situations or environmental cues. The course rejects the idea that positive habits appear effortlessly; instead, they need sustained practice, while negative habits can develop quickly through surroundings and peer influence.

The habit cycle is explained as a repeating chain in which a thought or stimulus leads to an action or reaction, followed by a reward or punishment. Repetition strengthens this pattern until it becomes deeply established and begins influencing character, performance and life outcomes. Examples from R.K. Narayan's The Guide and William Golding's Lord of the Flies illustrate the presence of both destructive impulses and the human capacity to choose better behaviour.

Dopamine is discussed as a brain chemical associated with pleasure and satisfaction. It can support productive behaviour when achievement itself becomes rewarding, but easy-pleasure cycles can also encourage unhealthy behavioural addictions. Another major concept is the Zeigarnik Effect: unfinished activities remain active in the mind, creating anxiety and consuming cognitive capacity. Starting a task can therefore reduce the psychological barrier to completion and create momentum.

The week closes with characteristics of effective people, including consistent hard work, willingness to take on additional responsibilities, maintaining a positive attitude, concentrating beyond the clock, and deliberately replacing unproductive routines with constructive alternatives.

Week 4: Active Listening & Professional Telephone Communication

Week 4 treats listening as an active communication skill rather than simply the physical ability to hear. Effective listening involves a seven-stage process: reception, selection, organization, assimilation, interpretation, evaluation and response. Learners are encouraged to recognize that speaking ability does not remove the need for strong listening skills.

The module identifies several barriers to attentive listening. These include physical noise, bodily discomfort, partial attention, emotional reactions and personal bias toward the speaker. Overcoming these barriers requires deliberate concentration, empathy and appropriate use of silence.

Telephone communication receives special attention because the absence of visual and non-verbal signals increases the possibility of misunderstanding. Professional calls should be handled with preparation, including keeping writing materials ready, speaking with a warm tone, maintaining a suitable pace and using verbal signals such as "I see" and "go on" to show attention.

The course also stresses clear pronunciation, especially when similar-sounding words may cause confusion, along with organized workspaces and appropriate methods for communicating unpleasant news. When dealing with angry, demanding or impatient callers, professionals should control voice modulation, remain courteous, avoid unnecessary technical jargon and anticipate potential issues before they become larger problems. These practices help telephone conversations become opportunities to demonstrate professionalism and build trust.

Week 5: Digital Behaviour, Mobile Courtesy & Netiquette

Week 5 examines how technology influences personality, behaviour and relationships. The discussion uses Donna Haraway's concept of the "cyborg" to explore the growing boundary between human beings and machines. It questions whether technology is remaining a useful tool or gradually becoming something that controls attention and behaviour. The lectures also raise concerns about excessive dependence on devices and reduced face-to-face interaction.

Mobile-phone overuse is associated with "moborg" behaviour and nomophobia, the fear or anxiety of being without a mobile phone. Frequent checking can replace direct human interaction with virtual validation. The recommended response is to establish device-free boundaries, keep technology in its proper role as a tool, silence phones during formal situations and prefer direct communication when it is more appropriate.

Professional digital communication is organized around five important soft-skill principles: planning, preparedness, persuasiveness, presentability and perseverance. The course points out that poor email practices—such as missing salutations, excessive use of capital letters, slang, mixed-language expressions and weak grammar—can create negative impressions in professional environments.

Netiquette is therefore treated as an essential professional responsibility. Digital messages may be saved, forwarded or viewed by others long after they are sent. Communicators should remember that a real person is behind every screen, select words carefully, convey emotion thoughtfully and maintain ethical standards to protect their digital footprint.

Week 6: Communication Process, Barriers, Filters & Non-Verbal Foundations

Week 6 studies communication as a deliberate process in which the sender aims to create a desired response in the receiver. Harold Lasswell's communication formula—who says what, to whom, through which channel and with what effect—provides a framework for understanding this process. Effective messages depend on clarity, conciseness, conviction, appropriate timing and empathy.

Communication can fail because both sender and receiver operate through psychological filters. Differences in frames of reference, encoding and decoding mistakes, and emotional interference may change the meaning of a message. In organizations, communication may move downward, upward or horizontally through written, oral and electronic channels.

The module highlights structural barriers such as long reporting chains, rigid hierarchies, unnecessary transfer points and limited workplace trust. These can distort information as it moves through an organization, sometimes producing outcomes that differ substantially from the original intention. Examples from corporate communication and product development show how a message can lose its intended meaning.

To reduce such problems, professionals should choose communication channels intelligently, address fears related to organizational change and use feedback to confirm shared understanding. The week concludes with an introduction to non-verbal communication and pre-assessment activities intended to identify existing knowledge and challenge common assumptions about body language.

Week 7: Body Language, Non-Verbal Signals & Professional Evaluation

Week 7 develops the study of non-verbal communication in greater depth. The course emphasizes the significant role of body language in emotional communication and notes that non-verbal signals often originate from subconscious feelings, making them difficult to manipulate consistently. Such signals may repeat, complement, contradict, regulate or replace spoken words.

The main areas of non-verbal communication include kinesics (movement and gestures), facial expressions, oculesics (eye behaviour), haptics (touch), proxemics (personal space), chronemics (time) and paralinguistics (tone, pitch and pauses). These cues can be understood through three psychological dimensions: immediacy, which relates to liking and closeness; arousal, which reflects responsiveness or excitement; and dominance, which concerns power and control.

Correct interpretation requires the “3 Cs”: context, clusters and changes from baseline behaviour. A single gesture should not be judged in isolation. Cultural differences and environmental conditions must also be considered before reaching conclusions about another person’s feelings.

The final part applies non-verbal principles to interviews and group discussions. Candidates are encouraged to be punctual, dress appropriately, use a confident handshake, maintain suitable eye contact and sit with composed posture. During group discussions, aggressive gestures, nervous movements and complete silence should be avoided. Calm participation, open-palm gestures, pleasant expressions and attentive nodding help communicate confidence and cooperation.

Week 8: Presentation Skills, Efficient Reading & Professional Integrity

The final week combines public speaking, reading strategies and professional ethics. Presentation anxiety is linked to fear of humiliation, unfamiliar situations and unexpected events. Stage fright can be reduced by mastering the subject, developing a confident self-image, converting nervous energy into enthusiasm and mentally visualizing a successful presentation.

Effective presenters should become familiar with the venue, welcome the audience positively and organize the talk around three or four clear central ideas. A strong presentation combines useful information with relatable human examples. Visual aids should remain simple and readable, using concise bullet points and clear fonts. Speakers should avoid obstructing the screen, keep backup copies when necessary, maintain an open posture and project their voice clearly rather than depending on a podium.

Reading efficiency is also addressed as a way of expanding perspective and strengthening vocabulary. Different techniques serve different purposes: previewing or sampling helps decide whether material is relevant; skimming identifies major themes; scanning locates particular information; clustering words can improve reading speed; and close reading supports detailed interpretation of philosophical or literary material.

The course ends by stressing that professional competence has little value without trust and ethical conduct. The golden rule—treating others as one would like to be treated—provides a simple ethical foundation. Circumstances do not create character; they reveal it. Long-term success depends on preserving integrity, avoiding impulsive betrayal and ensuring that interactions leave others with a sense of respect and value.

RESCUEAI

Project Ideation: Intelligent Emergency Response Platform

1. Problem Statement

During emergencies, people may be unable to type messages, explain their situation, or share their exact location. Traditional emergency calling may not provide responders with enough context or continuous location updates, which can delay assistance.

2. Proposed Solution

RescueAI is an intelligent emergency companion designed to enhance the phone's existing emergency dialer/SOS system rather than replace it. It uses AI to understand emergency reports, share essential information, and coordinate responses.

3. Key Features

One-Tap SOS	Initiates an alert and the phone's supported emergency-calling flow.
Multimedia SOS	Allows voice messages, photos, or short videos when typing is difficult.
AI Voice-to-Text	Transcribes speech, identifies the emergency type, and estimates priority.
Live Location	Shares GPS location with updates approximately every five minutes, subject to permissions and connectivity.
Responder Alerts	Sends structured emergency details to an authorized responder dashboard.
Response Tracking	Shows SOS Received → Responder Assigned → On the Way → Reached → Resolved.
Emergency Contacts	Notifies selected trusted contacts where supported.

4. What Makes It Unique?

Unlike a basic SOS button, RescueAI combines voice, visual evidence, AI-based emergency classification, periodic location updates, and response tracking in one platform.

5. How It Works

1. Activate SOS	The user taps SOS or uses a supported emergency trigger.
2. Collect details	The system gathers location and optional voice, photo, or video.
3. Analyse with AI	Speech is transcribed and the emergency type and priority are estimated.
4. Alert responders	A structured alert and location updates go to authorized responders.
5. Track response	The user sees status updates through resolution.

6. Technology Stack

Component	Suggested Technology
Frontend	HTML, CSS, JavaScript
Backend	Python Flask
Database	SQLite (initial prototype)
AI	Speech-to-text and text classification
Location	GPS / Android location services
Responder dashboard	Web application
Mobile integration	Android app using supported calling and permission APIs

7. Expected Impact

- Makes emergency reporting easier.
- Gives responders clearer information.
- Improves location visibility.
- Supports more transparent coordination.

8. Feasibility and Limitations

A hackathon prototype can demonstrate SOS alerts, voice transcription, simulated responder routing, location sharing, and status tracking. Real emergency-service integration requires authorization and agency coordination. The app must not claim help has been dispatched unless the receiving service confirms it. Background location updates depend on permissions, battery, and connectivity.

9. Project Objective

To develop an AI-powered emergency platform that makes reporting easier, gives responders useful context, and improves coordination from the first SOS alert through resolution.

> “Don’t replace the emergency dialer—make it intelligent, multimodal, and response-aware.”

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

Hanu Shree (She/Her)

First-Year Computer Science Engineering Student | Aspiring Software Developer | C & Python | Problem Solving | Hackathon Enthusiast

St. Joseph's College of Engineering, Chennai, Tamil Nadu, India

About

I am a first-year Computer Science Engineering student at St. Joseph's College of Engineering, Chennai, with a strong interest in software development and problem-solving.

Currently, I am building my foundation in C and Python while developing my skills in Data Structures and Algorithms (DSA) and SQL. I am interested in understanding how software systems are designed and solving problems through efficient and practical solutions.

I also enjoy exploring hackathons and project-based learning, where I can apply what I learn to real-world problems. My current goal is to strengthen my programming fundamentals, build meaningful projects, and continuously improve my technical skills.

Education

St. Joseph's College of Engineering

Bachelor of Engineering, Computer Science Engineering. Aug 2026 – Dec 2030

Activities and societies: coding activities, hackathons, technical events and college clubs, and exploring and developing new skills in programming and technology.

Skills

Python (Programming Language)

Connect

GitHub: github.com/hanushree-cse/hanushree_cse

LinkedIn: linkedin.com/in/hanu-shree-hanu-ab068642b

Part B: GitHub profile README

Hi, I'm Hanushree

CSE Student | Aspiring Software Engineer

I'm a Computer Science Engineering student passionate about learning how technology works and building useful projects.

About Me

- Currently pursuing Computer Science Engineering
- Currently learning C, Python, SQL & Data Structures
- Interested in Software & Backend Development
- Exploring Cloud & DevOps

- Curious about FinTech
- Always learning something new

2026 Learning Goals

- Strengthen programming fundamentals
- Improve problem-solving & DSA
- Learn Python properly
- Master SQL & DBMS
- Build real-world projects
- Learn Backend Development
- Explore Cloud & DevOps

Areas I'm Interested In

- Software Development
- Cloud & DevOps
- Backend Engineering

My Goal

To build strong programming fundamentals, create real-world projects, and grow into a skilled software engineer.

Tools & Technologies

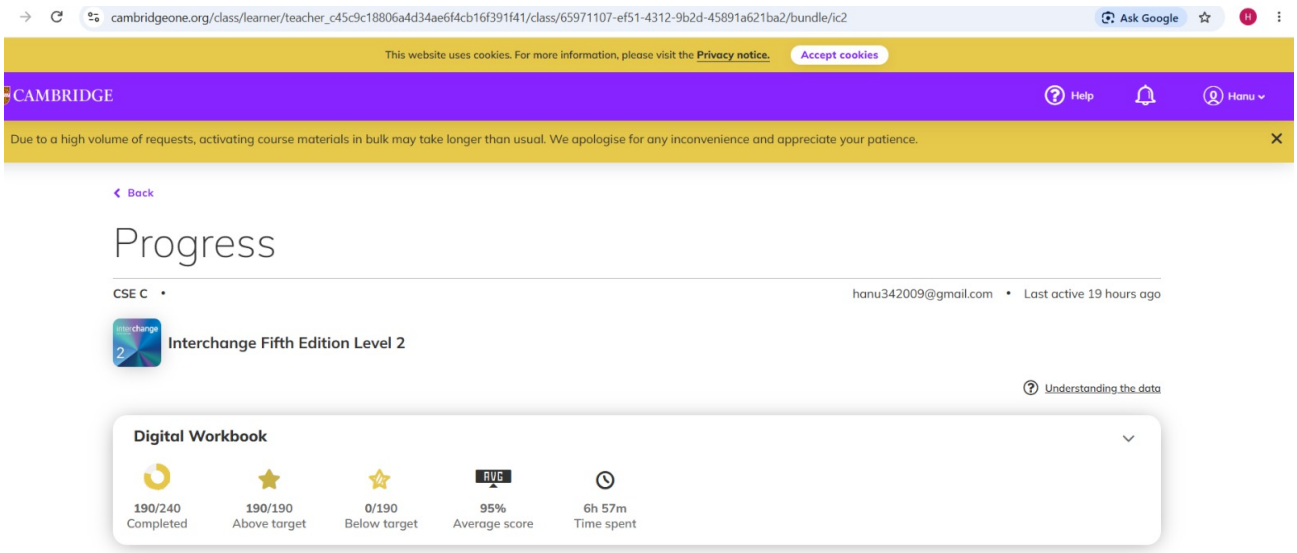
Domain	Technologies & Tools
Languages	C, Python
Database	SQL
Version Control	Git, GitHub
Web	HTML, CSS

Currently Working On

Improving my coding skills and building beginner-friendly projects.

Connect With Me

LinkedIn: https://www.linkedin.com/in/hanu-shree-hanu-ab068642b


cambridgeone.org/class/learner/teacher_c45c9c18806a4d34ae6f4cb16f391f41/class/65971107-ef51-4312-9b2d-45891a621ba2/bundle/ic2
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Understanding the data
Digital Workbook
190/240
Completed
190/190
Above target
0/190
Below target
AVG
95%
Average score
6h 57m
Time spent

4. CEFR level: B1 (solid B1)

- **Why B1:** You tell a connected story in sequence, you use a range of structures (relative clause, "After + -ing", "asked + object + to do"), and your vocabulary is accurate for the topic (convert, electrical signals, gyroscope, fingerprint). Errors exist, but the meaning is always clear.
- **Why not B2:** B2 needs consistent grammatical control and more varied sentence structures. You still have recurring slips (tense, agreement, verb forms) and few complex linking devices like "although," "which means," or "as a result."

ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) | OMR, Chennai - 600 119
Course Code: EN25CO9 — Professional English - I (Lab Integrated)
Model - I Practice Worksheet & Examination Answer Key

PART 1: MODEL – I PRACTICE WORKSHEET SOLUTIONS

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. Neither the project supervisor nor the lead programmers recognize the anomaly in the build script.
2. Each of the automated sorting arms requires recalibration after twelve hours of continuous duty.
3. Either the primary sensor or the auxiliary telemetry units transmit packet data to the ground station.
4. The consortium of aerospace contractors agrees to adhere to the revised tolerance levels.
5. Physics and material science form the core theoretical foundation for nanoscale fabrication.

Exercise 2: Simple Past Tense (Trial Reporting)

1. Last month, the metallurgy division conducted destructive fatigue testing on the titanium alloy shaft.
2. The field engineers deployed thirty IoT water-monitoring nodes across the river basin.
3. During the pressure overload trial, the relief valve opened at 150 psi as designed.
4. The research unit published their initial benchmark results on low-power RISC-V architectures.
5. In the 2025 dry run, the autonomous navigation algorithm detected all dynamic road obstacles successfully.

Exercise 3: Comparative Adjectives (Technical Metrics)

1. Gallium nitride (GaN) semiconductors operate at higher power efficiency than standard silicon transistors.
2. Solid-state batteries deliver a safer operational threshold under rapid discharge cycles than traditional lithium cells.
3. The revised compiler generates code that executes faster than the legacy unoptimized routine.
4. Brushless DC motors are considerably more durable than brushed equivalents due to reduced mechanical wear.
5. The liquid cooling loop maintained a lower core temperature than the dual fan-sink array.

Exercise 4: Modals of Obligation & Necessity

1. Every operator must wear anti-static wrist straps before handling open CMOS motherboards.
2. Hazardous chemical waste should be neutralized prior to disposal into the industrial drain network.
3. Field technicians have to log their entry time into the high-voltage testing bay without fail.
4. To maintain ISO certification, calibration logs must be audited biannually by an external inspector.
5. The test rig has to undergo an emergency stop inspection before initiating the high-speed turbine run.

Exercise 5: Modals of Capability & Possibility

1. With the upgraded DSP chip, the system can process 4K video streams in real time.
2. If thermal throttling occurs, the processing clock rate may drop by as much as 30%.
3. Advanced laser profilometry can measure surface roughness down to sub-micron tolerances.
4. Unshielded signal lines might pick up electromagnetic interference from nearby induction coils.
5. The modular chassis design could accommodate up to four secondary expansion cards.

Exercise 6: Relative Clauses

1. A civil engineer who received the national infrastructure award designed the suspension cable bridge.
2. The spectrometer, which is mounted inside the cleanroom vacuum chamber, analyzes infrared emission spectra.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller features ultra-low power sleep states, which makes it ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals)

1. If the bus voltage exceeds 24V, the zener diode will divert the surge to ground.
2. The watchdog timer will reset the CPU if the main thread fails to ping it within 500 ms.
3. If the operator engages the safety latch, the hydraulic press will not actuate.
4. If temperature levels rise above 90°C, the cooling fans will switch to maximum RPM.
5. The system will trigger an auditory alarm if the toxic gas detector registers more than 20 ppm.

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting)

1. If the backup generator had started immediately, the data centre servers would not have shut down.
2. The turbine bearing would not have seized if the maintenance crew had lubricated it as scheduled.
3. If the security team had patched the zero-day exploit earlier, the data breach would have been prevented.
4. The circuit breaker would have tripped safely if the overcurrent sensor had functioned properly.
5. If the telemetry link had remained stable, ground engineers would have corrected the satellite's orbital trajectory.

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. The aerodynamic team improved aircraft fuel economy by redesigning the winglet profile.
2. Always disconnect the power supply before replacing defective surface-mount components.
3. The facility was praised for reducing volatile organic chemical emissions by 40%.
4. Avoid cleaning precision optical lenses with abrasive industrial solvents.
5. The digital signal processor isolates target audio signals by filtering out background ambient noise.

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.

4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. The stepper motor rotates precisely in 1.8-degree angular increments.
2. The telemetry ground station periodically polls all deployed marine buoy sensors.
3. The thermal imaging camera registered the temperature drop rapidly during quenching.
4. Structural joints must be welded firmly to withstand heavy cyclic vibrational stress.
5. The autonomous drone traversed the confined pipeline corridor smoothly.

Exercise 13: Reported Speech (Interrogative Structures)

1. The quality manager asked the technician if the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer if he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. The procurement officer purchased several new machines and items of equipment for the workshop.
2. The laboratory generated a huge amount of test data during the wind tunnel trials.
3. We need three more software licenses to complete the finite element analysis on the model.
4. The field report provided numerous valuable pieces of information regarding the soil composition.
5. All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. Because the heat sink had poor thermal paste contact, heat dissipation was worse than expected.
2. The redesigned microstrip antenna achieved a better signal-to-noise ratio than the baseline patch.
3. The autonomous vehicle traveled 50 km farther on a single charge after the weight reduction overhaul.
4. Switching to asynchronous task queues resulted in less CPU idle time during peak batch processing.
5. We require further metallurgical investigation into why the stress cracks formed along the weld seam.

PART B: ADVANCED APPLICATION EXERCISES

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting)

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs)

1. Cloud storage offers significantly higher elastic scalability than on-premise physical servers.

Conversely, on-premise physical servers involve more expensive initial deployment costs than cloud infrastructure.

2. Fiber-optic cables deliver greater data transfer bandwidth than copper twisted-pair cables.

Additionally, fiber-optic lines are less susceptible to electromagnetic interference than traditional copper wiring.

3. Electric vehicles generate zero tailpipe emissions, making them environmentally cleaner than internal combustion engine vehicles.

However, electric vehicles currently feature a higher initial purchase price than conventional gas-powered cars.

4. Hydraulic actuators provide stronger raw force output than electromechanical actuators.

On the other hand, electromechanical actuators present a lower risk of fluid leakage than hydraulic systems.

5. Relational (SQL) databases maintain stricter schema rigidity than document-oriented (NoSQL) databases.

In contrast, NoSQL architectures achieve faster horizontal query throughput under dynamic workloads.

Exercise 3: Relative Clauses (Sentence Synthesis)

1. The algorithm which was patented by the aeronautics team analyzes microscopic cracks in turbine blades.
2. The lead chemical engineer who noticed an abnormal pressure spike in chamber two authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system utilizes an artificial neural network that predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting)

1. (a) If the operating temperature exceeds 85°C, the auxiliary cooling system will automatically engage to prevent thermal breakdown.

(b) If the operator had pressed the emergency shutdown switch immediately, the mechanical damage to the turbine assembly would have been completely avoided.

2. (a) If the supply voltage drops below the 200V threshold, the low-voltage disconnect relay will isolate the sensitive processing circuits.

(b) If the network administrator had enabled multi-factor authentication, unauthorized remote access to the database server would have been blocked.

3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the automated exhaust ventilation fans will trigger at high speed.

(b) If the mechanical team had checked the bearing lubrication schedule, the main shaft would not have seized during peak load testing.

4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will apply immediate emergency braking and rerun pathfinding logic.

(b) If the thermal sensor had registered the runaway reaction in time, the safety relief valve would have vented the chamber before over-pressurization.

5. (a) If the software receives an unauthenticated API token, it will terminate the active session and log an unauthorized access violation.

(b) If the civil inspectors had reviewed the core sample density tests, they would have identified the foundation structural flaws prior to construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.
2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. Emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech (Interrogative Technical Transformations)

1. The safety supervisor asked the intern if he/she had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he/she had disabled the firewall rule on the database server.
4. The client asked the project manager if the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he/she had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new software tools and machines in the CAD laboratory.

(b) The team collected plenty of useful information from the field test.

2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.

(b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.

3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.

(b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.

4. (a) All electrical hardware and wiring must comply with local safety legislation.

(b) The data scientist discovered several interesting insights after running the machine learning clustering.

5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.

(b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose & Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
(b) The automated script runs every midnight for archiving legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
(b) The technicians analyzed structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
(b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
(b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
(b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner & Operational Precision

1. (a) automatically | (b) flawlessly
2. (a) accurately | (b) instantly
3. (a) continuously | (b) cautiously
4. (a) consistently | (b) catastrophically
5. (a) periodically | (b) smoothly

Exercise 10: Modals of Capability, Obligation, and Prohibition

1. (a) can | (b) must not
2. (a) must | (b) may
3. (a) must not | (b) could
4. (a) should | (b) must
5. (a) can | (b) must not

PART 2: MODEL EXAM WORKSHEET (EN25C09)

PART - A SHORT ANSWERS (15 MARKS)

1. monitors
2. recorded
3. stronger
4. must
5. can
6. The laboratory developed a new testing device which measures material strength with high precision.
7. rises
8. would not have failed
9. The mechanics installed dual damper springs to isolate chassis resonance.
10. optimizing
11. 50 mL of hydrochloric acid was poured into the titration flask.
12. smoothly
13. The project supervisor asked the engineer if he/she had calibrated the pressure sensor before testing.
14. The laboratory purchased several new pieces of equipment for conducting material testing.
15. worse

PART - B SHORT APPLICATION ANSWERS (20 MARKS)

1. The engineer calibrated the pressure sensor and verified its accuracy.
2. (Portability): A laptop is significantly more portable than a desktop computer, allowing mobile computing.

(Screen Size): A desktop computer typically provides a larger display screen size than a standard laptop.

3. The engineer designed a new cooling system that reduces the temperature of industrial machines.
4. (a) If the battery voltage drops below the required level, the circuit will isolate the power supply automatically.

(b) If the technician had checked the connections earlier, the equipment would not have short-circuited.

5. The solution was heated to 80°C before testing.
6. The project manager asked the technician whether he/she had inspected the pressure valves before starting the compressor.
7. (a) The engineer received much feedback on the new design.

(b) The students received several useful pieces of advice from their teacher.

8. (a) The technician wears safety gloves to protect his hands.

(b) The cooling fan operates continuously to prevent the motor from overheating.

9. (a) efficiently | (b) carefully
10. (a) can | (b) must not

PART - C: LONG-FORM & ANALYTICAL WRITING (65 MARKS)

Question 26 (a): Eight Imperative Safety Instructions for Cyber Security

1. Use complex passwords containing uppercase letters, numbers, and symbols for all active accounts.
2. Enable multi-factor authentication (MFA) across all academic and personal access portals without fail.
3. Never click on unverified links or download unexpected attachments embedded in unsolicited emails.
4. Verify website domain names for valid HTTPS encryption before inputting credentials or personal data.
5. Update operating systems, firmware, and antivirus applications routinely to patch critical vulnerabilities.
6. Avoid conducting sensitive banking or academic transactions over unsecured public Wi-Fi access points.
7. Back up critical research data daily onto encrypted external drives or authorized cloud repositories.
8. Report suspicious system anomalies, data breach warnings, and phishing attacks immediately to network administrators.

Question 27 (a): Descriptive Essay — Preserving Traditional Folk Art Forms

Introduction: Traditional folk art forms like Therukoothu (street theatre) represent the vibrant cultural ethos of South India. Historically staged during seasonal harvest festivals, this dynamic art form combines classical music, theatrical dialogue, and intricate hand-crafted costumes to depict epic mythologies and convey timeless moral values to rural communities.

Origins & Traditional Practices: Rooted in ancient village rituals, Therukoothu was traditionally performed in open-air spaces under torchlight. Performers relied on vocal endurance, natural pigment cosmetics, and heavy timber props. It served not only as community entertainment but as a critical oral channel for preserving historical narratives across generations.

Modern Initiatives for Sustainability: In the contemporary era, modern initiatives are essential to keep this art form relevant for younger generations. Digital archiving, livestreaming performances, and conducting collegiate drama workshops bridge the cultural divide. Replacing toxic synthetic cosmetics with eco-friendly organic dyes aligns the heritage art with modern health and environmental standards.

Conclusion: By combining digital visibility with sustained institutional support, traditional folk arts can thrive as meaningful expressions of cultural identity.

Question 28 (a): Formal Letter of Request — Extension of Library Hours

From:

Student Representative, First-Year CSE

St. Joseph's College of Engineering

OMR, Chennai – 600119

Date: November 2, 2026

To:

The Chief Librarian

Central Library

St. Joseph's College of Engineering

OMR, Chennai – 600119

Subject: Request for extension of library working hours during semester examinations – Reg.

Respected Sir/Madam, I am writing on behalf of the first-year engineering students to formally request an extension of the library working hours in view of the upcoming end-semester examinations.

Currently, the library closes at 6:00 PM. We request you to extend operational hours until 8:30 PM on all working days for the next three weeks. Extended access to technical reference textbooks, quiet study spaces, and digital research databases will significantly assist students in thorough exam preparation.

We anticipate daily participation from approximately 80 to 100 students. Student volunteers will assist library staff in maintaining strict discipline and orderly management of study stations.

Thank you for considering our request.

Yours sincerely,

Indiresh LG

(Class Representative, CSE)

Question 29 (a): Analytical Essay — AI-Based Assessment vs. Teacher-Based Assessment

Evaluation Criteria AI-Based Assessment Teacher-Based Assessment

Feedback Speed Instantaneous automated evaluation Requires several days for manual scoring

Objectivity & Bias Strictly algorithmic; zero subjective bias Potential risk of subjective grader bias

Qualitative Depth Limited evaluation of original design thinking Deep evaluation of logic, creativity, & intent

Essay: Engineering assessment is evolving rapidly through automated digital tools. Comparing Artificial IntelligenceBased Assessment with traditional Teacher-Based Assessment highlights clear trade-offs in operational efficiency, objectivity, and qualitative feedback.

On one hand, AI systems deliver instantaneous results and evaluate massive student cohorts effortlessly. They eliminate subjective bias in objective testing and automated code verification. Conversely, teacher-based assessment offers deep qualitative evaluation. Human educators can judge original engineering design choices, evaluate partial logical reasoning, and appreciate creative approaches that algorithms misinterpret.

Ultimately, a balanced hybrid framework—utilizing AI for rapid objective screening while reserving human expertise for creative evaluation—presents the optimal path for engineering education.

Question 30 (a): Personal Email — First-Semester Experience

To: dynamic.friend@email.com

Subject: Catching up! My first-semester experience at St. Joseph's

Dear Rahul, I hope you are doing well! I wanted to drop a line and share how my first semester in Computer Science Engineering is progressing.

Transitioning to college life has been an enriching experience. I have formed great friendships, and we regularly collaborate on lab experiments and coding assignments. The hands-on sessions in the programming and physics labs have been engaging and practical.

Adapting to the fast-paced academic schedule was initial challenging, but regular group study sessions have kept us on track. With semester exams approaching next month, I am currently focused on revising lab manuals and past question papers.

How are things at your campus? Let's connect over a call this weekend!

Warm regards,

Indiresh